

Generation Interconnection System Impact Study Report

For Queue Project Q15-0531

Fair Oaks 345 kV

March 19, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0531. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Fair Oaks 345 kV substation in the Pennsylvania area. The request was received on October 28, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 253 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$160,455,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mtn View / Antrim 138 kV L
Contingency: Chambersburg / Fair Oaks 345 kV
Loading: 174.8% (Rating: 216 MVA)
DFAX: 11.0%
Pre-project loading: 98.1%
2. Facility: Mtn View / Lurich 138 kV
Contingency: Chambersburg / Fair Oaks Substation 345 kV
Loading: 174.1% (Rating: 223 MVA)
DFAX: 11.0%
Pre-project loading: 100.0%
3. Facility: Carrel 345-138 kV XFMR
Contingency: Fair Oaks / Chambersburg 345 kV
Loading: 140.8% (Rating: 692 MVA)
DFAX: 32.8%
Pre-project loading: 69.7%
4. Facility: Carrel / Lourich 138 kV
Contingency: Fair Oaks / Chambersburg 345 kV
Loading: 140.6% (Rating: 232 MVA)
DFAX: 11.0%
Pre-project loading: 69.4%
5. Facility: Otsego / E Waynesboro 138 kV
Contingency: Chambersburg 345-138 kV Autotransformer
Loading: 132.6% (Rating: 394 MVA)
DFAX: 7.6%
Pre-project loading: 103.5%
6. Facility: W Waynesboro / E Waynesboro 138 kV
Contingency: Chambersburg 345/138kV Transformer Bank
Loading: 128.2% (Rating: 434 MVA)

- DFAX: 7.6%
Pre-project loading: 101.8%
7. Facility: Carl Bailey / Otsego 138 kV
Contingency: Fort Humbug / Orchard Road 500 kV
Loading: 124.1% (Rating: 582 MVA)
DFAX: 9.0%
Pre-project loading: 100.9%
8. Facility: Fair Oaks / Chambersburg 345 kV L
Contingency: Carrel 345/138 kV Autotransformer
Loading: 120.4% (Rating: 1,200 MVA)
DFAX: 58.0%
Pre-project loading: 48.0%
9. Facility: Placentia / Carrel 138 kV L
Contingency: Fair Oaks / Chambersburg 345 kV Line
Loading: 114.9% (Rating: 203 MVA)
DFAX: 8.7%
Pre-project loading: 51.1%
10. Facility: Sunrise Ctr 500/345kV Transformer
Contingency: Cayucos Substation / Sunrise Center 500 kV
Loading: 114.8% (Rating: 593 MVA)
DFAX: 5.7%
Pre-project loading: 100.4%
11. Facility: Bon Aqua / Dry Gap 138 kV
Contingency: Orchard Rd / Ft Humbug 500 kV
Loading: 112.4% (Rating: 629 MVA)
DFAX: 9.0%
Pre-project loading: 91.0%
12. Facility: Hutchinson Plant 1 / W Kelso 138 kV
Contingency: Altavista Power Sta Substation / Rubidoux 138 kV Line
Loading: 112.4% (Rating: 631 MVA)
DFAX: 6.0%
Pre-project loading: 98.1%
13. Facility: Carl Bailey / Bon Aqua 138 kV
Contingency: Orchard Rd / Ft Humbug 500 kV L
Loading: 112.1% (Rating: 636 MVA)
DFAX: 9.0%
Pre-project loading: 90.9%
14. Facility: Dry Gap Substation / W Kelso 138 kV line
Contingency: Altavista Power Station / Rubidoux 138 kV Line
Loading: 111.9% (Rating: 644 MVA)
DFAX: 6.0%
Pre-project loading: 97.9%
15. Facility: Carrel / Fair Oaks 345 kV
Contingency: Chambersburg Substation / Fair Oaks 345 kV
Loading: 110.8% (Rating: 1,196 MVA)
DFAX: 88.0%
Pre-project loading: 0.4%
16. Facility: Pacentia / Mcconnellsburg 138 kV
Contingency: Chambersburg / Fair Oaks 345 kV
Loading: 108.9% (Rating: 211 MVA)
DFAX: 8.7%

Pre-project loading: 47.3%

17. Facility: N Lewiston / Chambersburg 138 kV

Contingency: E Waynesboro / W Waynesboro 138 kV Line

Loading: 108.2% (Rating: 395 MVA)

DFAX: 6.8%

Pre-project loading: 82.5%

18. Facility: Sunrise Ctr / Lutesville 345 kV

Contingency: Sunrise Center / Cayucos 500 kV

Loading: 107.0% (Rating: 1,629 MVA)

DFAX: 8.3%

Pre-project loading: 99.4%

19. Facility: Mcconnellsburg / Buffalo 138 kV

Contingency: Fair Oaks / Chambersburg 345 kV

Loading: 106.6% (Rating: 210 MVA)

DFAX: 8.7%

Pre-project loading: 44.9%

20. Facility: W Campus / Orchard Road 345 kV L

Contingency: Wes Campus / Fort Humbug 345 kV

Loading: 103.4% (Rating: 1,076 MVA)

DFAX: 5.6%

Pre-project loading: 95.6%

21. Facility: W Campus / Edes Tap 2 345 kV

Contingency: Talenenergy Martins Creek Llc Williamsport / Cayucos 500 kV Line

Loading: 102.5% (Rating: 1,639 MVA)

DFAX: 8.1%

Pre-project loading: 95.1%

22. Facility: Newton Station / Buffalo 138 kV

Contingency: Chambersburg / Fair Oaks 345 kV line

Loading: 102.5% (Rating: 215 MVA)

DFAX: 8.7%

Pre-project loading: 42.1%

23. Facility: Edes Tap 2 / Lutesville 345 kV

Contingency: Talenenergy Martins Crk Llc Williamsport Substation / Cayucos 500 kV

Loading: 100.1% (Rating: 1,642 MVA)

DFAX: 7.1%

Pre-project loading: 93.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mtn View / Antrim 138 kV	Rebuild line with higher-capacity conductor	\$8,381,000	66% (\$5,545,000)
Mountain View / Lourich 138 kV	Rebuild line with higher-capacity conductor	\$2,975,000	66% (\$1,969,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carrel 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$14,486,000	81% (\$11,674,000)
Carrel / Lourich 138 kV	Rebuild line with higher-capacity conductor	\$9,175,000	63% (\$5,753,000)
Otsego / E Waynesboro 138 kV	Rebuild line with higher-capacity conductor	\$4,837,000	46% (\$2,222,000)
E Waynesboro / W Waynesboro 138 kV line	Rebuild line with higher-capacity conductor	\$4,689,000	46% (\$2,154,000)
Otsego / Carl Bailey 138 kV	Rebuild line with higher-capacity conductor	\$7,015,000	60% (\$4,187,000)
Fair Oaks / Chambersburg 345 kV L	Rebuild line with higher-capacity conductor	\$29,085,000	78% (\$22,687,000)
Carrel / Placentia 138 kV	Rebuild line with higher-capacity conductor	\$3,318,000	94% (\$3,125,000)
Sunrise Tr 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$30,745,000	69% (\$21,249,000)
Bon Aqua / Dry Gap 138 kV	Rebuild line with higher-capacity conductor	\$5,811,000	60% (\$3,468,000)
W Kelso / Hutchinson Plant 1 138 kV	Rebuild line with higher-capacity conductor	\$9,940,000	57% (\$5,675,000)
Bon Aqua Substation / Carl Bailey 138 kV	Rebuild line with higher-capacity conductor	\$10,899,000	60% (\$6,505,000)
Dry Gap Substation / W Kelso 138 kV line	Rebuild line with higher-capacity conductor	\$5,441,000	57% (\$3,107,000)
Carrel / Fair Oaks 345 kV	Rebuild line with higher-capacity conductor	\$43,105,000	97% (\$41,746,000)
Pacentia / Mcconnellsburg 138 kV	Reconductor line	\$3,930,000	94% (\$3,701,000)
N Lewiston / Chambersburg 138 kV line	Reconductor line	\$1,065,000	68% (\$725,000)
Sunrise Ctr / Lutesville 345 kV	Reconductor line	\$15,071,000	33% (\$4,963,000)
Mcconnellsburg Substation / Buffalo 138 kV	Reconductor line	\$1,840,000	94% (\$1,733,000)
Orchard Rd / W Campus 345 kV	Reconductor line	\$5,959,000	36% (\$2,153,000)
W Campus / Edes Tap 2 345 kV	Reconductor line	\$8,051,000	24% (\$1,971,000)
Buffalo / Newton Station 138 kV	Reconductor line	\$1,650,000	94% (\$1,554,000)
Edes Tap 2 / Lutesville 345 kV	Reconductor line	\$11,198,000	23% (\$2,589,000)
	Total allocated to project		\$160,455,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.